

SCOTT EGBERT

scegbert.com | Longmont, Colorado | scottegbert219@gmail.com | 801.891.7122 | [linkedin/scott-egbert](https://www.linkedin.com/in/scott-egbert)

Optical systems engineer specializing in stabilized lasers, fiber and free-space optics, electro-optic modulation, opto-mechanical integration, system characterization, and troubleshooting.

SKILLS

Optical Systems: stabilized lasers, fiber and free-space optics, fiber amplifiers, frequency combs, nonlinear optics

Test and Characterization: optical test development, phase and frequency-noise metrology, spectrum analyzers, frequency counters

Controls and Integration: EOMs, AOMs, laser stabilization, Python GUIs, thermal control, alignment, opto-mechanics

CAD and Engineering: SolidWorks, CATIA, opto-mechanical design, compact packaging, prototype development and integration

RELEVANT EXPERIENCE

IMRA AMERICA, BOULDER RESEARCH LAB | RESEARCH SCIENTIST 2024 - present

- Owned system redesign and ruggedization of IMRA's passively stabilized dual-wavelength Brillouin laser (DWBL), reducing volume from 570 to 15 liters and converting a precision, laboratory-scale optical frequency reference into portable hardware currently operating across three continents without sacrificing performance.
- Integrated 3.3 THz compact DWBL prototype with a Kerr microresonator oscillator to divide down to a 300 GHz carrier with 135 attosecond RMS timing jitter from 1 kHz to 1 MHz (results accepted in Nature Photonics).
- Implemented and characterized an EOM-based optical frequency-division system using feed-forward noise cancellation to improve X-band synthesizer phase noise by up to 30 dB (Nature Communications).
- Developed control GUI and standard operating procedures for portable DWBL systems, enabling collaborators to independently operate and troubleshoot hardware.

UNIVERSITY OF COLORADO, BOULDER | POSTDOCTORAL AND GRADUATE RESEARCHER 2019 - 2024

- Developed and integrated the first GHz-repetition-rate mid-infrared dual-comb spectrometer, combining stabilized fiber combs and amplifiers, nonlinear frequency conversion, free-space beam delivery, detection, and analysis to resolve microsecond-scale, multi-species chemical kinetics in a shock tube (Optica).
- Led optical and mechanical redesign of the lab-scale GHz MIR DCS into the first portable MIR dual-comb spectrometer, reducing footprint by 80% for rack deployment; hired, trained, and mentored three interns.
- Developed an object-oriented Python library and GUI for H₂O spectral database optimization, reducing processing time from years to two weeks and database errors from 23% to 1%; results incorporated into HITRAN 2024.
- Independently built a new FPGA-based real-time phase-correction system from a NIST design, sourcing and assembling hardware and debugging software to optimize spectral database measurements.

SANDIA NATIONAL LABS | LASER SCIENTIST INTERN Summer 2018

- Analyzed Raman-scattering images of LN₂ and LH₂ leak plumes, comparing measured and modeled concentrations to validate cryogenic release behavior and inform fire-safety guidance (Sandia technical report).

BRIGHAM YOUNG UNIVERSITY, PROVO, UT | GRADUATE RESEARCHER 2017 - 2019

- Designed and tested a sapphire fiber-optic probe for 1500 K, 12-atm measurements in an industrial gas turbine combustor, achieving ± 15 K measurement repeatability (ASME Turbo Expo).

LOCKHEED MARTIN AERONAUTICS | AERONAUTICAL DESIGN ENGINEER INTERN Summer 2015 and 2016

- Evaluated design improvements to F-35 flaps and wing edges at Edwards AFB, coordinating with manufacturing and design teams and incorporating changes using Product Data Management software.

EDUCATION

PhD in Mechanical Engineering | UNIVERSITY OF COLORADO, BOULDER, CO May 2024

- Enabling High-Temperature Measurements with Frequency Comb Laser and Spectral Database Development
- Advised by Dr. Greg Rieker in the Precision Laser Diagnostics Lab, funded by the US Air Force Research Lab

MS in Mechanical Engineering | BRIGHAM YOUNG UNIVERSITY, PROVO, UT Aug 2019

- Pressurized Combustion Product Temperature Measurement Using Integrated Spectral Band Ratios
- Advised by Dr. Dale Tree, funded by Solar Turbines

BS in Mechanical Engineering (Magna Cum Laude) | BRIGHAM YOUNG UNIVERSITY, PROVO, UT June 2017

- Teaching Assistant for thermodynamics and optics courses, Research Assistant developing multi-user CAD

HONORS AND AWARDS

- International Microwave Symposium post-deadline paper and presentation, "Tunable microwave frequency synthesis with optically-derived spectral purity" (2026)
- University of Colorado Vogel Family Fellowship (2023-2024)
- Gordon Research Conference - Laser Diagnostics in Energy, invited student talk (2023)
- QUADMARTS Conference Bowling Champion (2023)
- 2nd place, Colorado Photonics Industry Association University Meeting Poster Competition (2022)

SELECTED PUBLICATIONS

- **S.C. Egbert**, B.M. Heffernan, J. Greenberg, W.F. McGrew, A. Rolland, "Attosecond-timing millimeter waves via Kerr optical frequency division", accepted in **Nature Photonics**, 2026.
- J. Greenberg, **S.C. Egbert**, W.F. McGrew, B.M. Heffernan, A. Rolland, "Tunable microwave frequency synthesis with optically-derived spectral purity", **Nature Communications**, 2026.
- I.E. Gordon, ... **S.C. Egbert**, ... G.B. Rieker, ..., "The HITRAN2024 molecular spectroscopic database," Journal of Quantitative Spectroscopy and Radiative Transfer, 2026.
- **S.C. Egbert**, J. Greenberg, B.M. Heffernan, W.F. McGrew, A. Rolland, "Dual-wavelength Brillouin lasers as compact opto-terahertz references for low-noise microwave synthesis," Optics Express, 2025.
- N. Hoghooghi, P. Chang, **S.C. Egbert**, et al., S.A. Diddams, P. Lynch, G.B. Rieker, "GHz repetition rate mid-infrared frequency comb spectroscopy of fast chemical reactions," **Optica**, 2024.
- **S.C. Egbert**, D. Zeltner, M. Rezasoltani, D.R. Tree, "High Pressure Optical Measurements of Temperature at Turbine Rotor Inlet Conditions," ASME Turbo Expo, 2020.
- Complete publication list on [Google Scholar](#).